Transhipment problem

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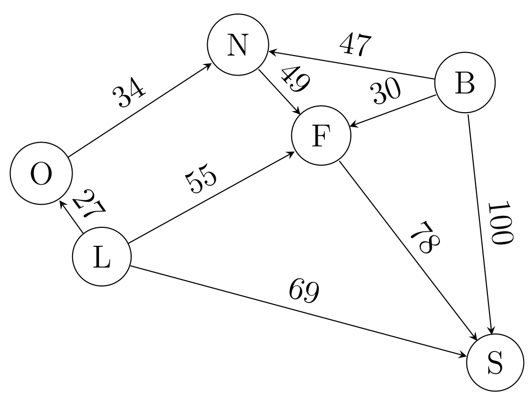
Practice quiz

A watch company in Switzerland has two factories in Lausanne and Bern. They supply 350 units, and 400 units per day respectively. Customers have a demand of 125 units per day in each of the following cities: Orbe (O), Neuchatel (N), Lausanne (L), Fribourg (F), Berne (B), Sion (S). The logistic system is designed in such a way that transportation between two cities is possible only in one direction, as illustrated in the figure below. The costs for transporting one unit are also reported in the following picture.



The transport from one city to another is limited to 200 units per day.

1. Write down a transhipment problem to minimize the total cost.

2. Transform it into a linear problem in standard form, such that the matrix A of the constraints has exactly two non zero entries in each column: 1 and -1.